

Bruno Grau Espinosa

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EDUCATION

University of Wyoming

Expected May 2027

College of Engineering and Physical Sciences: BS in Mechanical Engineering; *GPA:* 3.604/4.000

- *Honors:* Full-Ride Scholarship (Academic & Athletic); Tau Beta Pi Honors Engineering Society; CSCAA Scholar All-American ('25, '26); Diver of the Year ('24).
- *Relevant Coursework:* Manufacturing Processes, Machine Design, Physical Computing, System Dynamics, Mechanics of Materials, Thermodynamics, Fluid Dynamics, Transport Phenomena.

PROFESSIONAL EXPERIENCE

MASDEU ENGINYERS S.L., Barcelona, Spain

July 2026

Engineer Intern

- Developed an automated workflow that generated georeferenced engineering plans from Excel-defined coordinates and map scales, reducing manual effort in engineering document production.
- Implemented a document-processing tool that optimized construction documentation for AI processing, reducing token usage by approximately 90%.
- Enhanced an internal project management platform by automating project classifications and organizing project status, documentation, and pending tasks.

NOBA Ventures, Barcelona, Spain

June 2025-August 2025

Business Design Intern

- Prototyped and validated new business ideas for 5 clients (3 national, 2 international) through market research, competitive analysis, and smoke testing.
- Benchmarked 30+ global companies for an industrial machinery client and analyzed 2,000+ customer insights to guide product strategy for a leading European dairy producer.

University of Wyoming Department of Mechanical Engineering, Laramie, WY

February 2025-May 2025

Undergraduate Research Trainee

- Engaged in research with a team of international graduate students on frontal polymerization-based 3D printing of lunar composites.
- Prepared DCPD/lunar regolith composite specimens and performed tensile testing to evaluate the effects of material composition and printing parameters on mechanical performance.

PROJECTS

Composite Wall Heat Transfer Analysis, University of Wyoming

Spring 2026

- Calculated 1D heat transfer through a 10-section composite wall using thermal circuit analysis and identified an alternative insulation material that reduced heat loss by 52%.
- Validated results against a 2D SolidWorks thermal simulation, reconciling a 20-28% deviation by isolating multidimensional heat flow effects not captured in the 1D model.
- Projected \$200 in annual heating cost savings per structure from the insulation redesign, supported by an economic and environmental impact assessment.

Track-Following Autonomous Mobile Platform, University of Wyoming

Spring 2026

- Built and programmed a dual-motor Raspberry Pi robot with IR and ultrasonic sensors, achieving autonomous line-following across 3 distinct track geometries.
- Developed a multithreaded Python braking system that halted the platform within a 12-inch obstacle threshold and resumed motion automatically upon clearance.
- Implemented Event-based thread synchronization to coordinate concurrent sensor polling and motor control in real time.

LEADERSHIP EXPERIENCE & ACTIVITIES

University of Wyoming Varsity Swimming and Diving Team, Laramie, WY

August 2023-Present

NCAA Division I Student-Athlete

- Balanced 20+ hours of weekly Division I athletic commitments with a demanding engineering curriculum while mentoring teammates and coordinating with coaches and staff to support top three conference team and individual results.

University of Wyoming Student-Athlete Advisory Committee (SAAC), Laramie, WY

January 2025-Present

Swimming and Diving Team Representative

- Elected to represent 60+ Swimming & Diving student-athletes, advocating for team needs and facilitating communication with athletic department leadership on policies, initiatives, and student-athlete development.

SKILLS

- *Programming & Computing:* MATLAB, Python, Raspberry Pi, Microsoft Office, AI-Assisted Development.
- *Design:* SolidWorks (*Certified SOLIDWORKS Associate*).
- *Hardware and Prototyping:* Sensor Integration, Embedded Systems, Data Acquisition.
- *Language:* Spanish (Native), Catalan (Native), English (Professional Proficiency).